Process Concept

- An operating system executes a variety of programs:
  - Batch system – **jobs**
  - Time-shared systems – **user programs** or **tasks**
- Textbook uses the terms ***job*** and ***process*** almost interchangeably
- **Process** – a program in execution; an active program can be called a process.
- Process execution must progress in sequential fashion.

- Program is ***passive*** entity stored on disk (**executable file**), process is ***active***
  - Program becomes process when executable file loaded into memory
- Execution of program started via GUI mouse clicks, command line entry of its name, etc
- One program can be several processes
  - Consider multiple users executing the same program

Program	Process
It is a set of instructions that has been designed to complete a certain task.	It is an instance of a program that is being currently executed.
It is a passive entity.	It is an active entity.
It resides in the secondary memory of the system.	It is created when a program is in execution and is being loaded into the main memory.
It requires memory space to store instructions.	It requires resources such as CPU, memory address, I/O during its working.
It doesn't have a control block.	It has its own control block, which is known as Process Control Block.

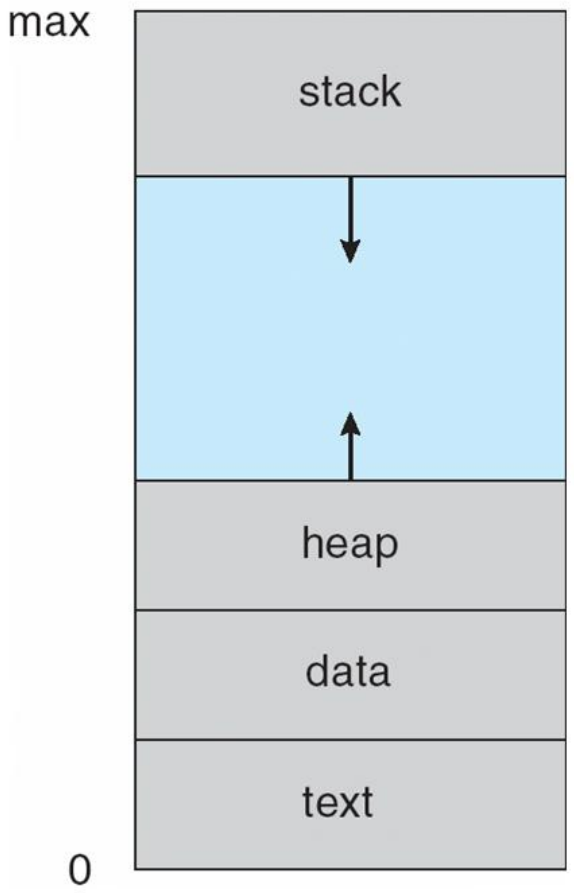
Process in Memory

✓ **Stack** containing temporary data
Function parameters, return addresses, local variables.

✓ **Heap** containing memory dynamically allocated during run time.

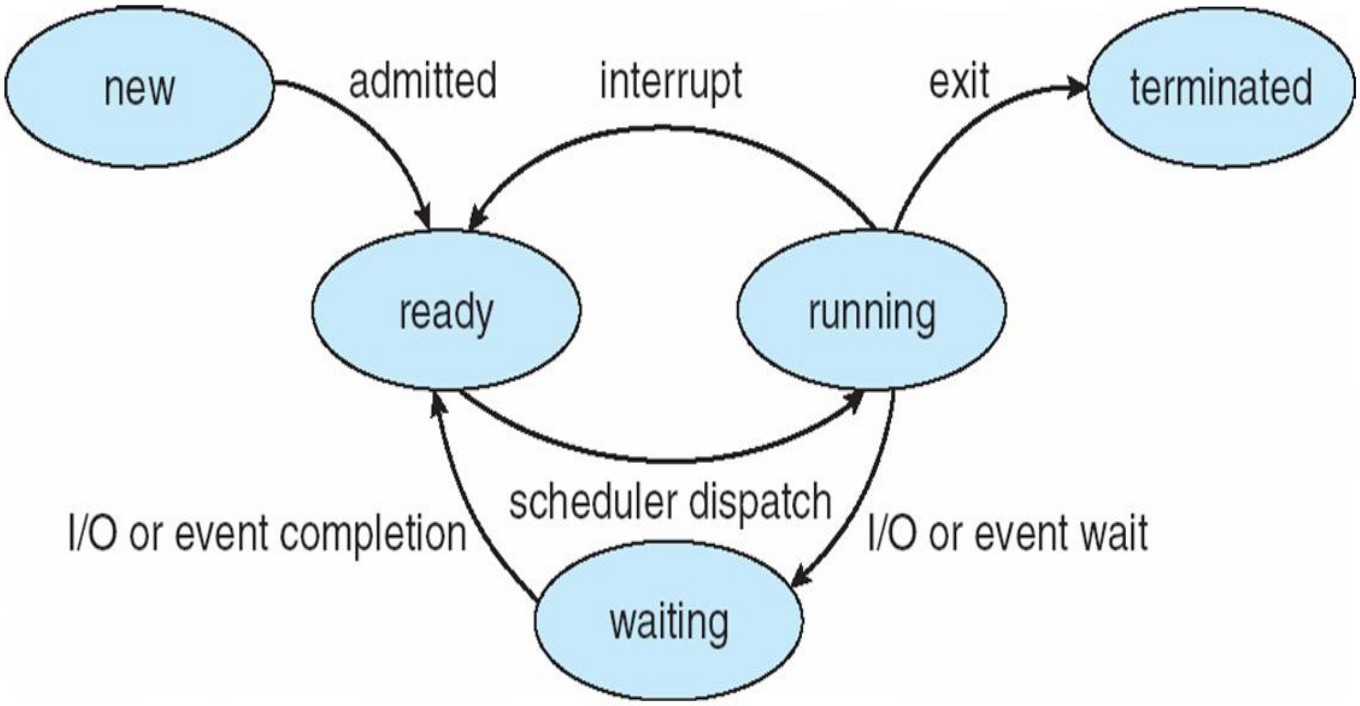
✓ **Data section** containing global and static variables.

✓ The program code, also called **text section**
Current activity including **program counter**, processor registers.



Process in Memory

Process State



- As a process executes, it changes **state**
  - **new**: The process is being created
  - **running**: Instructions are being executed
  - **waiting**: The process is waiting for some event to occur
  - **ready**: The process is waiting to be assigned to a processor
  - **terminated**: The process has finished execution

Process Control Block (PCB)

Information associated with each process (also called task control block)

- **Process ID -**

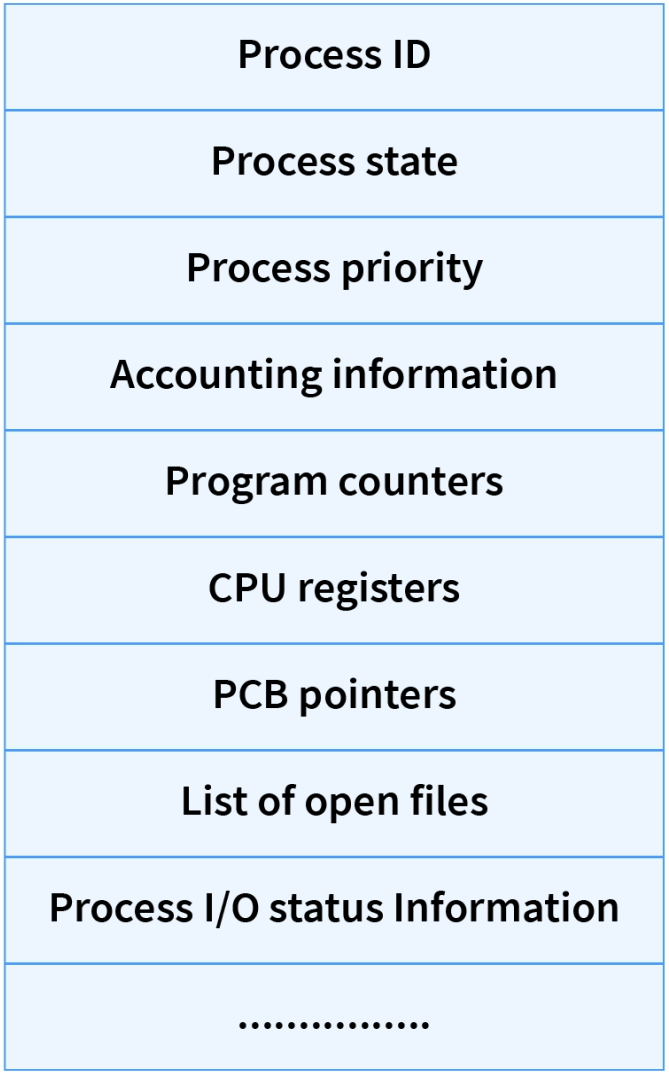
When a new process is created by the user, the operating system assigns a **unique ID.**

- **Process state –**

New, running, ready, waiting, etc.

- **Program counter –**

location of instruction to next execute.



- **Process Priority-**

Process priority is a numeric value that represents the priority of each process. The lesser the value, the greater the priority of that process.

- **CPU registers –**

contents of all process (centric registers).

- **PCB pointer-**

This field contains the address of the next PCB, which is in ready state. This maintain an easy control flow between parent processes and child processes.

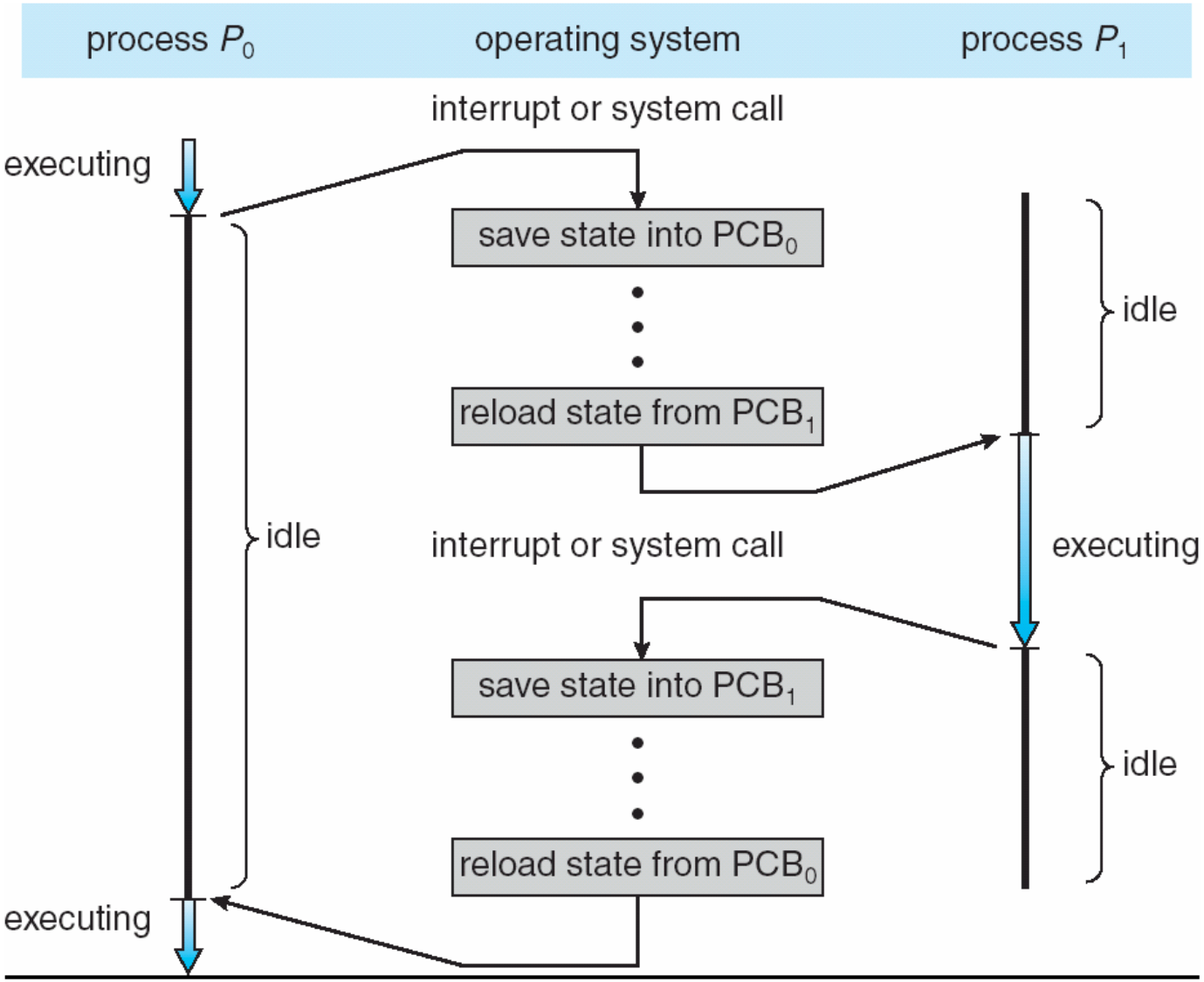
- I/O status information – I/O devices allocated to process, list of open files

Context switching

Context switching in an operating system (OS) refers to the process where the CPU switches from executing one process to another.

The steps involved in context switching are as follows −

- Save the context of the process that is currently running on the CPU. Update the process control block and other important fields.
- Move the process control block of the above process into the relevant queue, such as the ready queue, I/O queue, etc.
- Select a new process for execution.
- Update the process control block of the selected process. This includes updating the process state to running.
- Update the memory management data structures as required.
- Restore the context of the process that was previously running when it is loaded again on the processor. This is done by loading the previous values of the process control block and registers.


process P_0
operating system
process P_1
interrupt or system call
executing
save state into PCB_0
reload state from PCB_1
idle
idle
interrupt or system call
executing
save state into PCB_1
reload state from PCB_0
idle
executing

Process Scheduling

- Maximize CPU use is the objective of multiprogramming.
- Quickly switch processes onto CPU is the objective of time sharing.
- **To meet these objectives the Process scheduler** selects among available processes for next execution on CPU.
- Maintains **scheduling queues** of processes
  - **Job queue** – set of all processes in the system.
  - **Ready queue** – set of all processes residing in main memory, ready and waiting to execute.
  - **Device queues** – set of processes waiting for an I/O device
  - Processes migrate among the various queues

Representation of Process Scheduling

Queueing diagram represents queues, resources, flows

